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GEOGRAPHY · India: soils and natural vegetation

UNDERSTAND IT FIRST

What is it? Soil is the thin top layer of the land, made of **broken rock (minerals), humus (decayed plants and animals), water and air**. It forms very slowly from the **parent rock**, shaped by **climate, relief, plants and animals, and time**. **Natural vegetation** is the plant cover that grows **on its own**, without human planting; it depends mainly on **rainfall and temperature**, and on **altitude** in the hills. Soils decide **which crops grow where**; forests give timber, fuel, medicine and protect soil and water. Learn it as **how soil forms → the major soil types → soil erosion and conservation → forest types by rainfall → mangroves and forest cover → AP**.

1. Soils of India

Major soils of India

Soil	Formation and where	Nature and crops
Alluvial	Laid down by **rivers**; the **largest** group (about **40%** of India); **Indo-Gangetic plains** and the **east-coast deltas** (Mahanadi, Godavari, Krishna, Kaveri). **Khadar** = new, fertile, flooded each year; **bhangar** = old, higher, with lime nodules (**kankar**).	Rich in **potash and lime**, poor in **nitrogen and phosphorus**; very fertile; **rice, wheat, sugarcane, jute**.
Black (regur)	From **Deccan lava (basalt)**; **Maharashtra**, Gujarat, MP, and parts of **Karnataka, Telangana and AP**.	Clayey; **holds moisture**; swells when wet and **cracks when dry** (**'self-ploughing'**); rich in **lime, iron, magnesium, potash**, poor in **N, P and humus**; **cotton** (hence 'black cotton soil'), jowar, tobacco, chillies.
Red and yellow	From old **granite and gneiss** under low rain; **Tamil Nadu, AP, Karnataka, Odisha, Chhattisgarh, Jharkhand**; the second largest group.	**Red** from **iron oxide**; **yellow** when the iron is hydrated; sandy, porous, poor in N, P, humus; with irrigation and fertiliser: **groundnut, millets, pulses, tobacco**.
Laterite	From the Latin **'later' = brick**; formed by **heavy rain and high heat**, which **leach** away silica and lime; **Western Ghats, Kerala, Karnataka, Odisha, NE hills**, parts of the **Eastern Ghats**.	Rich in **iron and aluminium**, poor in fertility; hardens in air, so cut as **building bricks**; with manure: **cashew, tea, coffee, rubber**.
Arid, saline,	**Arid**: Rajasthan (sandy, salty; kankar). **Saline/alkaline** (**reh, kallar, usar**): dry areas and over-irrigated land (Punjab, Haryana). **Peaty/marshy**: heavy rain and	Saline soils are treated with **gypsum**; peaty soils are rich in **humus**; arid soils grow crops with irrigation (Indira Gandhi canal).

peaty, forest	poor drainage (**Kerala's kari soil**, Sundarbans). **Forest/mountain**: Himalayan slopes.	

- **Classification: ICAR** classifies India's soils into **8 major groups** (the five above plus arid, saline and peaty/forest). The **US (USDA) taxonomy** is also used (e.g. **Inceptisols, Entisols, Alfisols, Vertisols** = black soils).
- **Soil erosion** is the removal of topsoil by **water** (**sheet, rill and gully** erosion; gullies make **ravines** like the **Chambal badlands**) and **wind** (Rajasthan). Causes: **deforestation, overgrazing, shifting cultivation (jhum), wrong ploughing**. **Conservation: contour ploughing, terrace farming** (hills), **strip cropping, shelter belts** (rows of trees against wind), check dams, afforestation. The **Soil Health Card scheme (2015)** tests soil nutrients and advises fertiliser use; **5 December** is **World Soil Day**.

2. Natural vegetation

Natural vegetation and rainfall

- **Tropical evergreen (rain) forests** (rain **over 200 cm**): dense, tall, layered; trees shed leaves at different times so the forest is **always green**; **Western Ghats, NE India, Andaman and Nicobar**; ebony, mahogany, rosewood, rubber. **Tropical deciduous (monsoon) forests** (**70–200 cm**): the **most widespread** type in India; trees **shed leaves** in the dry season; **moist** (sal, teak, bamboo; eastern India, Eastern Ghats) and **dry** (teak, sandalwood, red sanders); **Eastern Ghats of AP** (Nallamala, Seshachalam, Paderu–Araku hills) are mostly deciduous.
- **Thorn forests and scrub** (**below 70 cm**): Rajasthan, Gujarat, rain-shadow parts of the Deccan; acacia, babul, khejri, cacti, with long roots and thick, small leaves. **Montane forests** change with **height** like climate zones: wet temperate (oak, chestnut) at 1,000–2,000 m, then **pine, deodar, fir**, then **alpine** meadows and shrubs; **sholas** (patches of evergreen forest with grassland) in the **Nilgiris, Anamalai and Palani** hills. **Mangroves (tidal forests)** grow in **salty, muddy deltas**; **breathing roots (pneumatophores)**; **Sundarbans** (sundari tree, Royal Bengal tiger), **Bhitarkanika** (Odisha), **Coringa** (Godavari delta), Krishna delta mangroves, Pichavaram (TN).
- **Forest cover (India State of Forest Report 2023, released Dec 2024): forest and tree cover** about **25.2%** of India's area (forest cover about **21.8%**, tree cover about **3.4%**); largest forest area: **Madhya Pradesh**, then Arunachal Pradesh and Chhattisgarh; highest share of area under forest: **Mizoram** (among States) and **Lakshadweep** (among UTs); mangrove cover about **4,992 sq km**, with **West Bengal** having the largest share. The **National Forest Policy 1988** aims for **33%** of the land under forest (**two-thirds** in the hills). The report is published every **two years** by the **Forest Survey of India (Dehradun)**. **AP's** red sanders (**Pterocarpus santalinus**) is **endemic** to the Seshachalam–Lankamala–Palakonda hills.

EXAM TRAP

Soil = minerals + **humus** + water + air; factors: parent rock, climate, relief, organisms, time; **ICAR 8 groups**; **alluvial ~40%, largest** (khadar new, bhangar old with **kankar**; potash rich, N, P poor; rice, wheat); **black/regur** (Deccan basalt; Maharashtra; moisture, **self-ploughing** cracks; lime, iron, Mg rich; N, P, humus poor; **cotton**; Vertisols); **red and yellow** (granite, gneiss; **iron oxide**; TN, AP, Karnataka, Odisha;

groundnut, millets); **laterite** ('later' = brick; **leaching** in heavy rain and heat; iron, aluminium; bricks; **cashew, tea, coffee**); arid (Rajasthan); **saline: reh, kallar, usar → gypsum**; **peaty: Kerala kari, Sundarbans**; erosion: sheet, rill, **gully (Chambal ravines)**, wind; conservation: **contour ploughing, terracing, strip cropping, shelter belts**; **Soil Health Card 2015**; **World Soil Day 5 Dec**; forests: **evergreen >200 cm** (W Ghats, NE, Andamans; ebony, mahogany), **deciduous 70–200 cm (most widespread**; teak, sal, sandalwood; **AP Eastern Ghats**), **thorn <70 cm** (babul, khejri), **montane** (oak → pine, deodar → alpine; **sholas**), **mangroves** (pneumatophores; **Sundarbans, Bhitarkanika, Coringa**, Pichavaram); **ISFR 2023: ~25.2% forest + tree; MP largest area; Mizoram highest %; mangroves ~4,992 km², WB largest**; **FSI Dehradun, every 2 years**; **Forest Policy 1988: 33%**; **red sanders endemic to Seshachalam**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Alluvial soil is the most widespread soil group in India.

2. Black soil develops deep cracks in the dry season and is called 'self-ploughing'.

3. Laterite soil is rich in nitrogen and humus.

4. The red colour of red soil is due to iron oxide.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. Laterite soil is leached by heavy rain and is poor in nitrogen, phosphorus and humus.

PRACTICE

Match List-I (forest type) with List-II (typical trees):

List-I (forest type)	List-II (typical trees)
A. Tropical evergreen	1. Teak and sal
B. Tropical deciduous	2. Sundari
C. Thorn and scrub	3. Ebony and mahogany
D. Mangrove	4. Khejri and babul

(1) A → 1 | B → 3 | C → 4 | D → 2

(2) A → 3 B → 4 C → 1 D → 2

(3) A → 3 B → 1 C → 2 D → 4

(4) A → 3 B → 1 C → 4 D → 2

Answer: (4) Evergreen: ebony, mahogany; deciduous: teak, sal; thorn: khejri, babul; mangrove: sundari.

PRACTICE

Which one of the following statements is NOT correct?

A. Madhya Pradesh has the highest percentage of its area under forest.

B. Tropical deciduous forests are the most widespread forests in India.

C. The National Forest Policy 1988 aims at 33% of land under forest.

D. Coringa mangroves are in the Godavari delta.

Answer: A. Madhya Pradesh has the largest forest area; Mizoram has the highest percentage of its area under forest.

PRACTICE

Assertion (A): Laterite soils are generally poor in fertility.
Reason (R): Heavy rainfall and high temperature leach away silica and nutrients, leaving mainly iron and aluminium oxides.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Both are true, and leaching is exactly why laterite is infertile.

PRACTICE

A farmer on the Deccan plateau has dark, clayey soil that becomes sticky when wet, cracks when dry and holds moisture for long. Which crop is this soil best known for?

A. Tea

B. Jute

C. Cotton

D. Rubber

Answer: C. This is black (regur) soil, known as black cotton soil.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write how soil forms and the ICAR soil groups with origin, nutrients and crops. 2. Write soil erosion types and conservation methods. 3. Write the five forest types with rainfall and trees, mangroves, ISFR 2023 figures and AP's forests.

ECONOMY · RBI: functions, monetary policy and credit control

UNDERSTAND IT FIRST

What is it? The **Reserve Bank of India (RBI)** is India's **central bank**: the bank of the Government and of all other banks. Its main job is to keep **prices stable while supporting growth**. It does this through **monetary policy**: changing the **cost and amount of money and credit** in the economy. When prices rise too fast, it makes loans **costlier and scarcer** (tight money); when growth is weak, it makes them **cheaper** (easy money). Learn it as **history → functions → the Monetary Policy Committee → quantitative tools → qualitative tools → current rates**.

1. History and functions

- Set up on the advice of the **Hilton Young Commission (Royal Commission on Indian Currency and Finance, 1926)** under the **RBI Act, 1934**; began work on **1 April 1935** in **Calcutta**, moved to **Bombay (1937)**; started as a **private shareholders' bank** and was **nationalised on 1 January 1949**. First Governor **Sir Osborne Smith**; first Indian Governor **C. D. Deshmukh**; present Governor **Sanjay Malhotra** (26th, from **December 2024**). It is run by a **Central Board of Directors**.

Functions of the RBI

- Besides these, the RBI does **developmental work**: it helped build **NABARD (1982)**, **SIDBI, EXIM Bank**, the **Deposit Insurance and Credit Guarantee Corporation (DICGC)** (deposits insured up to **₹5 lakh** per depositor per bank), and promotes **financial inclusion** and **digital payments** (the **Payment and Settlement Systems Act, 2007**; **NPCI**, which runs UPI, was set up with RBI support). It also manages the **foreign exchange reserves** under **FEMA 1999**.

2. The Monetary Policy Committee

- On the **Urjit Patel Committee (2014)** report, India adopted **flexible inflation targeting** (Monetary Policy Framework Agreement, **February 2015**). The **RBI Act was amended in 2016** to create a **statutory Monetary Policy Committee (MPC)**: **six members**: **three from the RBI** (Governor as chair, a Deputy Governor, one officer) and **three external members** appointed by the **Government** for **four years**. Each has one vote; the **Governor has a casting vote** in a tie. It meets **at least four times a year** (in practice six). The **Government**, in consultation with the RBI, sets the **inflation target** every **five years**: **CPI inflation 4%**, with a band of **2% to 6%**. If inflation stays outside the band for **three quarters in a row**, the RBI must **explain in writing** to the Government.

3. Tools of monetary policy (credit control)

Tools of monetary policy

Tool	Meaning
Repo rate	Rate at which the RBI gives **short-term (overnight) loans to banks** against **government securities**, with an agreement to buy them back ('repurchase'). The **main policy rate**; set by the **MPC**.
SDF and reverse repo	The **Standing Deposit Facility (April 2022)** lets banks **park surplus money with the RBI without giving securities**; it is the **floor** of the rate corridor (repo minus 0.25%). The **reverse repo** (RBI borrowing against securities) still exists but is no longer the floor.
MSF and bank rate	The **Marginal Standing Facility (2011)** gives banks **emergency overnight money** at a **penalty** rate (repo plus 0.25%), even by **dipping into their SLR** securities; it is the **ceiling** of the corridor. The **bank rate** is the rate for longer loans without repurchase; now **kept equal to the MSF** and used for **penalties**.
CRR and SLR	**Cash Reserve Ratio**: share of a bank's **net demand and time liabilities (NDTL)** kept as **cash with the RBI**; earns **no interest**. **Statutory Liquidity Ratio** (under the **Banking Regulation Act 1949**): share of NDTL the bank keeps **itself** in **cash, gold or approved government securities**. Raising either **reduces lending**.
Open market operations (OMO)	The RBI **sells** government securities to **absorb** money (fights inflation) and **buys** them to **inject** money. Also **variable rate repo/reverse repo auctions** and **forex swaps** manage day-to-day liquidity under the **Liquidity Adjustment Facility (LAF, 2000)**.
Qualitative (selective) tools	Control **where** credit goes, not how much: **margin requirements** (share of a loan the borrower must bring), **credit rationing** (ceilings), **regulation of consumer credit**, **moral suasion** (persuasion, advice), **direct action** (penalties on erring banks), **priority sector lending** (40% of credit for agriculture, small units, weaker sections).

- **Current settings (check the latest):** the MPC **raised the repo rate by 0.25% to 5.50%** at its **October 2026** meeting and changed its stance from 'neutral' to **'calibrated tightening'**; **SDF 5.25%**, **MSF and bank rate**

5.75%. The **CRR** was cut to **3%** in 2025 (phased, Sept–Nov 2025) and the **SLR** is **18%**. (Before this, the repo rate had come down from **6.50%** in steps through 2025.)

EXAM TRAP

Hilton Young 1926 → RBI Act 1934 → 1 April 1935 (Calcutta; Bombay 1937) → nationalised 1 Jan 1949; Osborne Smith, **C. D. Deshmukh**; **Sanjay Malhotra (26th, Dec 2024)**; functions: **bank of issue, banker to Govt (public debt, WMA), bankers' bank (lender of last resort), monetary authority, regulator, forex (FEMA 1999)**, development (NABARD, DICGC **₹5 lakh**, PSS Act 2007, NPCI); **Urjit Patel 2014**; **framework Feb 2015**; **2016 amendment: MPC of 6 (3 RBI + 3 Govt, 4 years; Governor casting vote; ≥4 meetings)**; **target CPI 4% ± 2 set by Govt for 5 years**; miss 3 quarters → written report; **repo** (main rate, against G-secs); **SDF 2022 = floor (repo − 0.25)**; **MSF 2011 = ceiling (repo + 0.25), can dip into SLR**; **bank rate = MSF**; **CRR (cash with RBI, no interest), SLR (BR Act 1949; cash, gold, G-secs with bank)** on **NDTL**; **OMO: sell = absorb, buy = inject**; **LAF 2000**; qualitative: **margin, rationing, consumer credit, moral suasion, direct action, priority sector 40%**; tight vs easy money; **Oct 2026: repo 5.50% (hike), SDF 5.25, MSF 5.75, stance 'calibrated tightening'; CRR 3%, SLR 18%**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. The RBI began operations on 1 April 1935 on the recommendation of the Hilton Young Commission.

2. The RBI was nationalised in 1949.

3. The Monetary Policy Committee has six members and the Governor has a casting vote.

4. The RBI alone fixes the inflation target without consulting the Government.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. The Government sets the inflation target in consultation with the RBI.

PRACTICE

Match List-I (instrument) with List-II (meaning):

List-I (instrument)	List-II (meaning)
A. Repo rate	1. Banks park surplus funds with the RBI without giving securities

B. Standing Deposit Facility	**2.** Share of deposits kept as cash with the RBI
C. Cash Reserve Ratio	**3.** Rate for overnight loans to banks against government securities
D. Marginal Standing Facility	**4.** Emergency overnight borrowing at a penalty rate, even from SLR holdings

(1) A → 3 B → 1 C → 2 D → 4

(2) A → 1 B → 3 C → 2 D → 4

(3) A → 3 B → 2 C → 1 D → 4

(4) A → 4 B → 1 C → 2 D → 3

Answer: (1) Repo: overnight loans against G-secs; SDF: deposit without collateral; CRR: cash with RBI; MSF: penalty-rate emergency borrowing.

PRACTICE

Which one of the following statements is NOT correct?

A. The SDF is the floor of the RBI's interest-rate corridor.

B. Banks earn no interest on CRR balances.

C. To absorb excess money, the RBI sells government securities.

D. Moral suasion is a quantitative tool of credit control.

Answer: D. Moral suasion is a qualitative (selective) tool: persuasion, not a rate or ratio.

PRACTICE

Assertion (A): To control inflation, the RBI may sell government securities in the open market.
Reason (R): The RBI acts as banker to the Government.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but selling securities controls inflation because buyers pay money to the RBI, which leaves less money with banks and the public; being the Government's banker does not explain this.

PRACTICE

A bank's net demand and time liabilities are ₹1,000 crore. If the CRR is 3% and the SLR is 18%, how much must it keep as cash with the RBI?

A. ₹30 crore

B. ₹180 crore

C. ₹210 crore

D. ₹300 crore

Answer: A. Only the CRR is kept as cash with the RBI: 3% of 1,000 = ₹30 crore. The SLR (₹180 crore) is held by the bank itself.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the RBI's history and its functions. 2. Write the MPC: law, members, voting, meetings and inflation target. 3. Write each quantitative and qualitative tool with its effect, and the latest rates.

SCIENCE · International environmental initiatives; India's role; sustainable development

UNDERSTAND IT FIRST

What is it? Since pollution, biodiversity loss and climate change cross borders, countries sign **conventions** (treaties) and **protocols** (detailed agreements under a convention) to act together. **Sustainable development** is the idea that ties them: growth that meets **today's needs without harming the ability of future generations to meet theirs**. Day 33 covered the climate treaties (UNFCCC, Kyoto, Paris); today covers **the big summits → sustainable development and the SDGs → biodiversity, chemical and ozone treaties → wetlands and wildlife → India's own global initiatives**.

Milestones of global environmental action

1. Summits, sustainable development and the SDGs

- **Stockholm (1972)**: the **UN Conference on the Human Environment**, the first global environment meeting; created **UNEP** (HQ **Nairobi**); **5 June** became **World Environment Day**; Indira Gandhi spoke of **poverty as the worst polluter**. The **Brundtland Commission** report **'Our Common Future' (1987)** defined **sustainable**

development; its **three pillars** are **economic growth, social equity and environmental protection** (people, planet, prosperity).

- **Rio Earth Summit (1992; UN Conference on Environment and Development)**: **Agenda 21** (action plan for the 21st century), the **Rio Declaration** (27 principles), **Forest Principles**, and two conventions opened for signature: the **UNFCCC** and the **Convention on Biological Diversity (CBD)**; the **UN Convention to Combat Desertification (UNCCD)** followed in **1994** (India hosted its **COP14 in 2019**). **Rio+20 (2012)**, 'The Future We Want', started the process for the **SDGs**.
- The **Millennium Development Goals (MDGs, 2000)**: **8 goals** for **2015**. The **Sustainable Development Goals (SDGs)**, adopted in **September 2015** as **Agenda 2030**: **17 goals, 169 targets**, for **all countries**, with the promise to **'leave no one behind'** (e.g. Goal 1 no poverty, 2 zero hunger, 6 clean water, 7 clean energy, 13 climate action, 14 life below water, 15 life on land). In India, **NITI Aayog** coordinates the SDGs and publishes the **SDG India Index** (first **2018**; in the **2023–24** edition **Kerala and Uttarakhand** topped; check the latest edition). **MoSPI** tracks the national indicators.

2. Other key conventions and protocols

Area	Agreements
Biodiversity	**CBD (1992)**: conservation, sustainable use, fair sharing of benefits. **Cartagena Protocol (2000)**: **biosafety** of **GM (living modified) organisms**. **Nagoya Protocol (2010)**: **access and benefit-sharing (ABS)** from genetic resources. **Kunming–Montreal Global Biodiversity Framework (2022)**: **'30x30'**: protect **30% of land and sea by 2030**. India: **Biological Diversity Act 2002** (**National Biodiversity Authority, Chennai**); India hosted **CBD COP11 (Hyderabad, 2012)**.
Chemicals and waste	**Basel (1989)**: control of **cross-border movement of hazardous wastes**. **Rotterdam (1998)**: **prior informed consent (PIC)** before trading dangerous chemicals and pesticides. **Stockholm (2001)**: eliminate **persistent organic pollutants (POPs)** such as DDT, dioxins ('dirty dozen'). **Minamata (2013)**: **mercury** (named after the mercury poisoning at Minamata, Japan).
Ozone	**Vienna Convention (1985)** → **Montreal Protocol (1987)** to phase out **CFCs, halons** and other ozone-depleting substances; the most successful treaty, ratified by **every UN member**. **Kigali Amendment (2016)** to **phase down HFCs**, which do **not** harm ozone but are powerful **greenhouse gases**; India's schedule: freeze by **2028**, **85% cut by 2047**.
Wetlands and wildlife	**Ramsar Convention (1971, Iran)**: **wetlands of international importance**; India joined in **1982**; first sites **Chilika and Keoladeo (1981)**; India now has **over 90 Ramsar sites** (the number keeps rising; check); **Kolleru** (AP) is one. **CITES (1973)**: controls **trade in endangered plants and animals** (Appendix I, II, III). **CMS / Bonn Convention (1979)**: **migratory species**; India hosted **CMS COP13 (Gandhinagar, 2020)**.

3. India's role and initiatives

- India argues for **equity** and **'common but differentiated responsibilities'**: rich countries polluted first and must give **finance and technology**; India's **per-capita emissions** are well below the world average. India's global initiatives: the **International Solar Alliance (ISA)**, launched by **India and France** at **COP21 Paris**

(2015), HQ **Gurugram**, the first international body headquartered in India; **'One Sun One World One Grid' (OSOWOG)**; the **Coalition for Disaster Resilient Infrastructure (CDRI, 2019)**, HQ **New Delhi**; **Mission LiFE** (**Lifestyle for Environment**, announced at **COP26, 2021**; launched **2022**): mindful use instead of wasteful consumption.

- At the **G20 New Delhi summit (September 2023)** India launched the **Global Biofuels Alliance** and the **Green Development Pact**; the **Green Credit Programme (2023)** rewards voluntary actions such as **tree planting and water conservation** with tradable **green credits**; the **International Big Cat Alliance (IBCA)**, for **seven big cats** (tiger, lion, leopard, snow leopard, cheetah, jaguar, puma), was launched in **2023** (HQ in India). At home: **Environment (Protection) Act 1986**, the **National Green Tribunal (2010)**, and schemes like **Nagar Van Yojana** and **'Ek Ped Maa Ke Naam' (2024)**.

EXAM TRAP

Stockholm 1972 (UNEP, Nairobi; 5 June); **Brundtland 'Our Common Future' 1987**: SD = needs of today without compromising future generations; **3 pillars: economy, society, environment**; **Rio 1992: Agenda 21, Rio Declaration, Forest Principles, UNFCCC, CBD; UNCCD 1994 (India COP14 2019)**; **Rio+20 2012**; **MDGs 2000: 8 goals, 2015**; **SDGs Sept 2015: 17 goals, 169 targets, 2030, 'leave no one behind'**; **NITI Aayog SDG India Index (2018; 2023–24 top Kerala, Uttarakhand)**; **CBD → Cartagena 2000 (biosafety, GMOs), Nagoya 2010 (ABS), Kunming–Montreal 2022 (30x30)**; **BD Act 2002, NBA Chennai; COP11 Hyderabad 2012**; **Basel 1989 (hazardous waste), Rotterdam 1998 (PIC), Stockholm 2001 (POPs), Minamata 2013 (mercury)**; **Vienna 1985 → Montreal 1987 (universal) → Kigali 2016 (HFCs; India freeze 2028, 85% by 2047)**; **Ramsar 1971 (India 1982; Chilika, Keoladeo first; 90+ sites; Kolleru)**; **CITES 1973 (trade)**; **CMS 1979 (Bonn; India COP13 2020)**; **CBDR, per-capita**; **ISA 2015 (India + France; Gurugram)**; **OSOWOG**; **CDRI 2019 (New Delhi)**; **LiFE 2021/2022**; **G20 2023: Global Biofuels Alliance, Green Development Pact**; **Green Credit 2023**; **IBCA 2023 (7 big cats)**; **EPA 1986, NGT 2010**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:
1. The Brundtland Report (1987) defined sustainable development.
2. Agenda 21 and the Convention on Biological Diversity came out of the Rio Earth Summit, 1992.
3. The SDGs consist of 17 goals to be achieved by 2030.
4. The Millennium Development Goals were 8 goals with a 2015 deadline.
Which of the above are correct?

A. 1 and 3 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: D. All four statements are correct.

PRACTICE

Match List-I (convention) with List-II (subject):

List-I (convention)	List-II (subject)
A. Basel Convention	1. Mercury
B. Rotterdam Convention	2. Persistent organic pollutants
C. Stockholm Convention (2001)	3. Prior informed consent for trade in hazardous chemicals
D. Minamata Convention	4. Cross-border movement of hazardous wastes

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 3 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (2) Basel: hazardous waste movement; Rotterdam: PIC; Stockholm: POPs; Minamata: mercury.

PRACTICE

Which one of the following statements is NOT correct?

A. The International Solar Alliance was launched by India and France in 2015.

B. The CDRI has its headquarters in New Delhi.

C. The International Solar Alliance has its headquarters in Paris.

D. The Global Biofuels Alliance was launched during India's G20 presidency.

Answer: C. The ISA's headquarters is at Gurugram, India.

PRACTICE

Assertion (A): The Kigali Amendment to the Montreal Protocol aims to phase down HFCs.

Reason (R): HFCs are major ozone-depleting substances.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: HFCs do not deplete ozone; they are controlled because they are powerful greenhouse gases.

PRACTICE

A State wants a large lake, important for migratory water birds, to be listed as a wetland of international importance. Under which convention is this done?

A. CITES

B. Ramsar Convention

C. Basel Convention

D. Nagoya Protocol

Answer: B. Wetlands of international importance are listed under the Ramsar Convention (1971); Kolleru and Chilika are examples.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write Stockholm, Brundtland, Rio 1992, Rio+20, the MDGs and SDGs, and the SDG India Index. 2. Write the biodiversity, chemicals, ozone, wetland and wildlife treaties with years and subjects. 3. Write India's stand and its global initiatives (ISA, CDRI, LiFE, Biofuels Alliance, Green Credit, IBCA).

NOTEBOOK PAGE · Day 34: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Soils and natural vegetation

Area	Write
Soils	Soil formation factors; ICAR's 8 groups; for each: origin, where, nutrients rich/poor, crops; soil erosion types and conservation; Soil Health Card

Vegetation	Five forest types with rainfall, regions and trees; mangroves; ISFR 2023 figures; Forest Policy 1988 (33%); AP forests (Eastern Ghats, Seshachalam red sanders, Coringa)

ECONOMY: RBI and monetary policy

- History (Hilton Young, 1934 Act, 1935, 1949); five functions; MPC (2016 amendment, members, inflation target); policy rates and the corridor; CRR, SLR, OMO; qualitative tools; tight vs easy money; worked CRR/SLR sum; latest rates (check).

SCIENCE: Global initiatives and sustainable development

Write: Stockholm, Brundtland definition and the three pillars, Rio 1992 outcomes, Rio+20, MDGs and SDGs, SDG India Index; biodiversity (CBD, Cartagena, Nagoya, Kunming-Montreal 30x30); chemicals (Basel, Rotterdam, Stockholm, Minamata); ozone (Vienna, Montreal, Kigali); Ramsar, CITES, CMS; India's initiatives (ISA, CDRI, LiFE, Global Biofuel Alliance, Big Cat Alliance, Green Credit).